RESEARCH CONCLUSION / 8 OCTOBER 2026

What can we conclude about Quiroga?

Our reconstruction of the printed adaptive update achieves high fidelity, but has not met our joint accuracy/TP gate within the tested budget. This is not a general failure verdict on Quiroga's method.

Evidence: two noiseless, rank-one, two-qubit calibration channels; two fixed TP penalties per channel; 100,000 printed updates each. All four runs completed normally, but 0/4 passed. These are small-scale diagnostics, not an n=8 result.

Metric	Observed range	Our gate	Outcome
Process fidelity	99.529% – 99.870%	≥ 99%	4/4 pass
Data-fit RMSE	1.045e-03 – 1.753e-03	≤ 1e-6	0/4 pass
Normalized TP error	2.205e-03 – 8.673e-03	≤ 1e-3	0/4 pass

What this means. It is not just a TP problem: RMSE is also about 1,000–1,750 times above our threshold. High fidelity alone therefore gives a more favorable impression than the stricter joint gate.

What we have narrowed down. Switching to a normalized Gell-Mann initializer changed Quiroga's endpoints but did not rescue any run. Armijo factored GD passes on both channels at the same fixed-penalty setting used by two of the printed-update runs. Thus the data and fixed-penalty formulation can support recovery here; optimization/step-size behavior is the leading explanation, not a uniquely established cause. Armijo remains a separate control, not Quiroga's published algorithm.

What we cannot claim. The paper gives the adaptive step only up to proportionality, and not all experimental normalization/initialization choices are pinned down [1]. We have not established an exact reproduction, impossibility of convergence, or failure at higher qubit counts. Nor is a final FRAMES victory established: FRAMES passes only 1/2 cases, narrowly missing the other.

Practical conclusion. Retain these as audited finite-budget results of the declared reconstruction. Before using Quiroga as a definitive best-for-best baseline, resolve or explicitly calibrate the step multiplier and normalization conventions, tune its fixed penalty independently of FRAMES, and validate frozen choices on unseen channels. No new experiment was launched for this note.

[1] Quiroga & Kyrillidis, arXiv:2312.01311v1, Sections III-B and V.

Evidence: audit_complete.json; source 966ef74; Ruche array 2038514; all eight archives audited.

Detailed paired results: QPT_GELL_MANN_DIAGNOSTIC_20261007.pdf